

KINECT CAMERAMAN

Problem

The Microsoft Kinect is a powerful device that allows for natural interaction with technology.



However, the limited field of view confines users into a small area.

Challenges

Many Kinect applications require a direct, perpendicular view of the user.

Most robots follow objects by rotating, causing a skewed view. Staying in front of the user at all times requires holonomic motion - movement in any direction without the need to rotate.

Holonomic robots are either too costly or too clunky to control in any useful way.

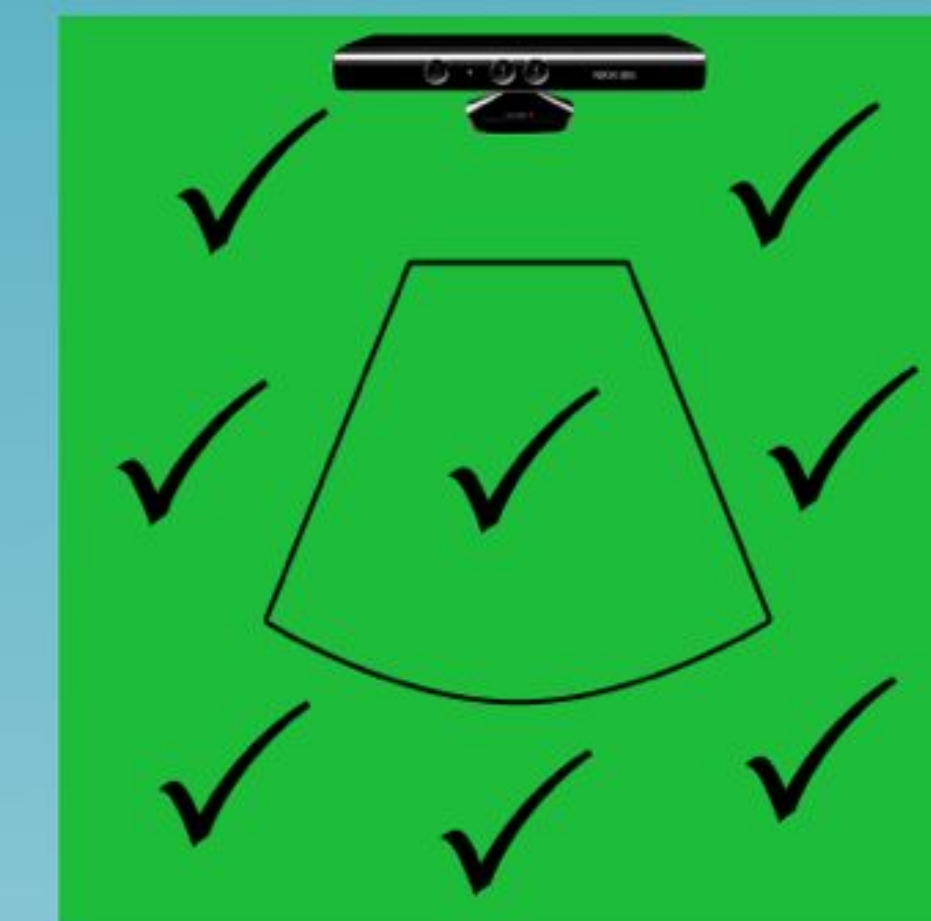
How do you stay in front of the user in a cost-effective and reliable way?

Solution

Through readily accessible hardware, we developed a robot that emulates holonomic motion.

By mounting the Kinect on a servo that counteracts all rotations of the robot, the Kinect is able to always face the same direction.

An inertial measurement unit (IMU) is included on-board to maintain a more precise heading.



Microsoft Kinect



iRobot Create



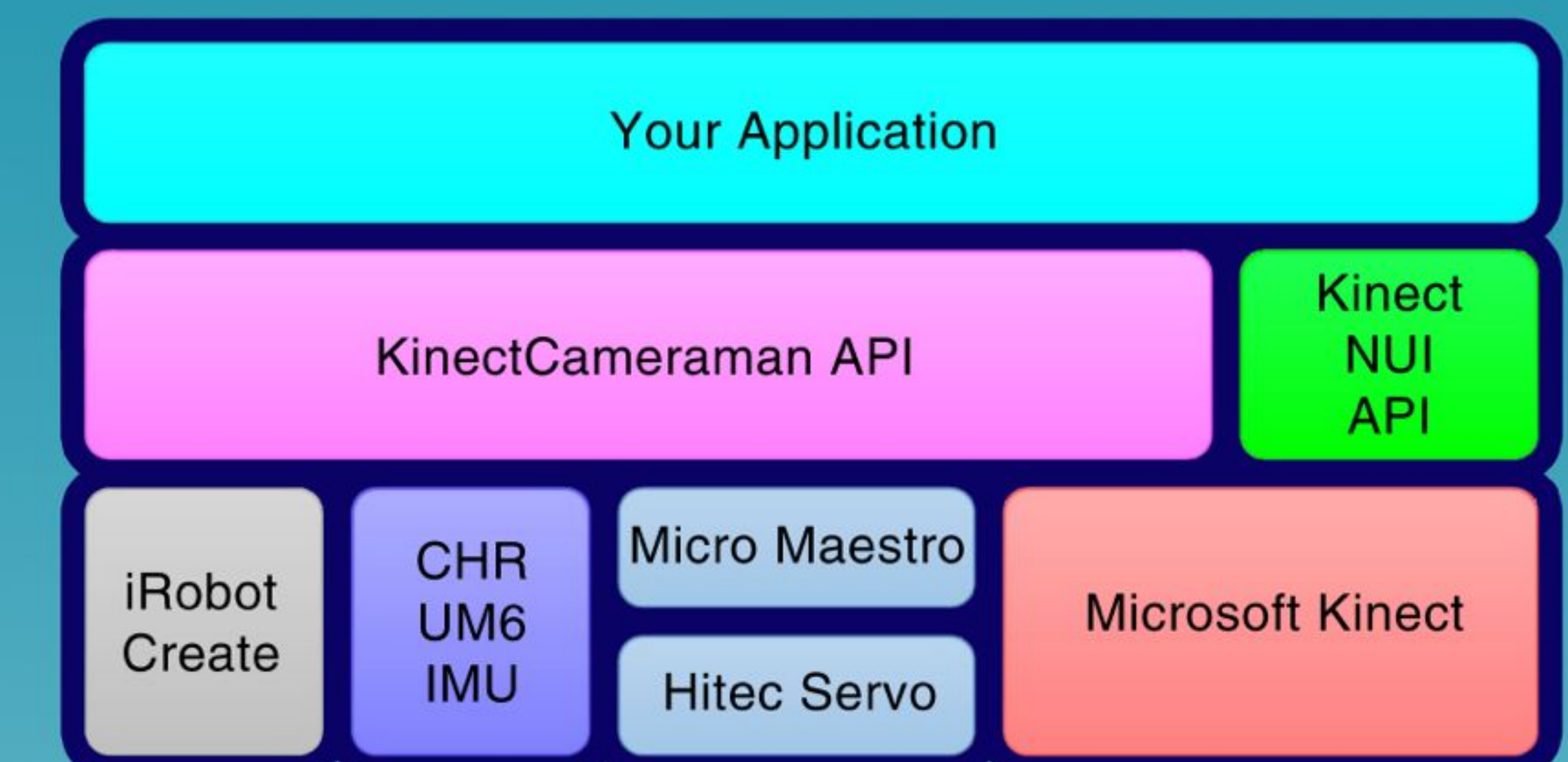
Micro Maestro
Servo Controller



Hitec Sail
Winch Servo



CH Robotics
IMU



Use Cases

- Telepresence (Skype)
- Games
- Virtual Environment Exploration
- Home Security
- Pet Care
- Webcasts
- Virtual Tours



Kenneth
Bedolla



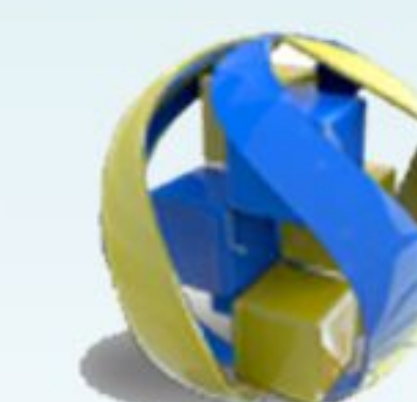
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